



1. Name of the medicinal product

APRITANT IV

Fosaprepitant Dimethylglumine Lyophilized Powder for solution for infusion 150 mg/vial

2. Quality and Quantitative composition:

Each vial contains 245.3 mg fosaprepitant dimethylglumine equivalent to 150 mg fosaprepitant.

3. Pharmaceutical form

White to off white lyophilized cake or powder .

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Fosaprepitant for injection, in combination with other antiemetic agents, is indicated for the prevention of acute and delayed nausea and vomiting associated with initial and repeat courses of:

- highly emetogenic cancer chemotherapy including high-dose cisplatin (see DOSAGE AND ADMINISTRATION)
- moderately emetogenic cancer chemotherapy (see DOSAGE AND ADMINISTRATION).

4.2 Posology and method of administration

APRITANT IV 150 mg is administered on Day 1 as an infusion over 20 – 30 minutes initiated approximately 30 minutes prior to chemotherapy. APRITANT IV 150 mg should be administered in conjunction with a corticosteroid and a 5-HT3 antagonist as specified in the tables below. The package insert for the co-administered 5-HT3 antagonist must be consulted prior to initiation of treatment with APRITANT IV 150 mg.

Recommended dosing for the prevention of nausea and vomiting associated with highly emetogenic cancer chemotherapy:

	Day 1	Day 2	Day 3	Day 4
APRITANT IV	150 mg IV	none	none	none
Dexamethasone**	12 mg orally	8 mg orally	8 mg orally bid	8 mg orally bid
5-HT ₃ antagonist	See the package insert for the selected 5-HT ₃ antagonist for the appropriate dosing information.	none	none	none

**Dexamethasone should be administered 30 minutes prior to chemotherapy treatment on Day 1 and in the morning on Days 2 through 4. Dexamethasone should also be administered in the evenings on Days 3 and 4. The dose of dexamethasone accounts for drug interactions.

Recommended dosing for the prevention of nausea and vomiting associated with moderately emetogenic cancer chemotherapy:

	Day 1
APRITANT IV	150 mg IV
Dexamethasone**	12 mg orally
5-HT ₃ antagonist	See the package insert for the selected 5-HT ₃ antagonist for appropriate dosing information.

**Dexamethasone should be administered 30 minutes prior to chemotherapy treatment on Day 1. The dose of dexamethasone accounts for drug interactions.

Preparation of APRITANT IV for Injection 150 mg

1. Inject 5 ml saline into the vial. Assure that saline is added to the vial along the vial wall in order to prevent foaming. Swirl the vial gently. Avoid shaking and jettling saline into the vial.
2. Prepare an infusion bag filled with 145 ml of saline.
3. Withdraw the entire volume from the vial and transfer it into an infusion bag containing 145 ml of saline to yield a total volume of 150 ml. Gently invert the bag 2-3 times.

The reconstituted final drug solution is stable for 24 hours at ambient room temperature (at or below 25°C). Parenteral drug products should be inspected visually for particulate matter and discoloration before administration whenever solution and container permit.

APRITANT IV is incompatible with any solutions containing divalent cations (e.g., Ca²⁺, Mg²⁺), including Hartman's and Lactated Ringer's Solution. APRITANT IV must not be reconstituted or mixed with solutions for which physical and chemical compatibility have not been established.

GENERAL INFORMATION

See DRUG INTERACTIONS for additional information on the administration of APRITANT IV with corticosteroids. Refer to the full prescribing information for coadministered antiemetic agents. No dosage adjustment is necessary based on age, gender, race or Body Mass Index (BMI). No dosage adjustment is necessary for patients with severe renal insufficiency (creatinine clearance <30 mL/min) or for patients with end stage renal disease undergoing hemodialysis. No dosage adjustment is necessary for patients with mild to moderate hepatic insufficiency (Child-Pugh score 5 to 9). There are no clinical data in patients with severe hepatic insufficiency (Child-Pugh score >9).

Instructions for Use

Physicians should instruct their patients to read the patient package insert before starting therapy with APRITANT IV and to reread it each time the prescription is renewed.

Patients should follow the physician's instructions for the APRITANT IV regimen. For the prevention of CINV, patients can be given a single dose of APRITANT IV 150 mg as an infusion over 20– 30 minutes, 30 minutes prior to chemotherapy on Day 1.

Advise patients to seek medical attention if they experience new or worsening signs or symptoms of an infusion site reaction, such as erythema, edema, pain, necrosis, vasculitis, or thrombophlebitis at or near the infusion site (see PRECAUTIONS).

APRITANT IV may interact with some drugs; therefore, patients should be advised to report to their doctor the use of any other prescription, non-prescription medication or herbal products.

Patients on chronic warfarin therapy should be instructed to have their clotting status closely monitored in the 2-week period, particularly at 7 to 10 days, following initiation of fosaprepitant with each chemotherapy cycle.

Concomitant administration of APRITANT IV may reduce the efficacy of hormonal contraceptives. Patients should be advised to use alternative or back-up methods of contraception during treatment with and for 1 month following administration of APRITANT IV.

Incompatibilities

APRITANT IV is incompatible with any solutions containing divalent cations (e.g., Ca²⁺, Mg²⁺), including Hartman's and lactated Ringer's solutions.

APRITANT IV must be reconstituted and then diluted prior to administration.

Preparation of APRITANT IV 150 mg for intravenous administration:

1. Inject 5 ml sodium chloride 9 mg/ml (0.9 %) solution for injection into the vial. Assure that sodium chloride 9 mg/ml (0.9 %) solution for injection is added to the vial along the vial wall in order to prevent foaming. Swirl the vial gently. Avoid shaking and jettling sodium chloride 9 mg/ml (0.9 %) solution for injection into the vial.
2. Prepare an infusion bag filled with 145 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection (for example, by removing 105 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection from a 250 ml sodium chloride 9 mg/ml (0.9 %) solution for injection infusion bag).
3. Withdraw the entire volume from the vial and transfer it into an infusion bag containing 145 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection to yield a total volume of 150 ml and final concentration of 1 mg/ml. Gently invert the bag 2-3 times.
4. Determine the volume to be administered from this prepared infusion bag, based on the recommended dose (see section Posology and method of administration).

Adults

The entire volume of the prepared infusion bag (150 ml) should be administered.

Paediatrics

In patients 12 years and older, the volume to be administered is calculated as follows:

- Volume to administer (ml) equals the recommended dose (mg)

In patients 6 months to less than 12 years, the volume to be administered is calculated as follows:

- Volume to administer (ml) = recommended dose (mg/kg) x weight (kg)

o Note: Do not exceed maximum doses (see section Posology and method of administration).

5. If necessary, for volumes less than 150 ml, the calculated volume can be transferred to an appropriate size bag or syringe prior to administration by infusion.

The appearance of the reconstituted solution is the same as the appearance of the diluent.

The reconstituted and diluted medicinal product should be inspected visually for particulate matter and discoloration before administration.

Discard any remaining solution and waste material. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

The medicinal product must not be reconstituted or mixed with solutions for which physical and chemical compatibility has not been established (see Incompatibilities)

Method of administration

APRITANT IV 150 mg should be administered intravenously.

4.3 Contraindications

Fosaprepitant is contraindicated in patients:

- who are hypersensitive to any component of the product Hypersensitivity reactions including anaphylactic reactions, flushing, erythema, and dyspnea have been reported.
- taking pimozone. Inhibition of CYP3A4 by apreptant, the active moiety, could result in elevated plasma concentrations of this drug, which is a CYP3A4 substrate, potentially causing serious or life- threatening reactions, such as QT prolongation, a known adverse reaction of pimozone.

4.4 Special warnings and precautions for use

Clinically Significant CYP3A4 Drug Interactions

Fosaprepitant, a prodrug of apreptant, is a weak inhibitor of CYP3A4, and apreptant is a substrate, inhibitor, and inducer of CYP3A4.

- Use of fosaprepitant with other drugs that are CYP3A4 substrates, may result in increased plasma concentration of the concomitant drug.
- Use of pimozone with fosaprepitant is contraindicated due to the risk of significantly increased plasma concentrations of pimozone, potentially resulting in prolongation of the QT interval, a known adverse reaction of pimozone.
- Use of Fosaprepitant with strong or moderate CYP3A4 inhibitors (e.g., ketoconazole, diltiazem) may increase plasma concentrations of apreptant and result in an increased risk of adverse reactions related to fosaprepitant.
- Use of fosaprepitant with strong CYP3A4 inducers (e.g., rifampin) may result in a reduction in apreptant plasma concentrations and decreased efficacy of fosaprepitant.

Hypersensitivity Reactions

Serious hypersensitivity reactions, including anaphylaxis and anaphylactic shock, during or soon after infusion of fosaprepitant have occurred. Symptoms including flushing, erythema, dyspnea, hypotension and syncope have been reported.

Monitor patients during and after infusion. If hypersensitivity reactions occur, discontinue the infusion and administer appropriate medical therapy. Do not reintitiate fosaprepitant in patients who experience these symptoms with previous use.

Infusion Site Reaction

Infusion site reactions (ISRs) have been reported with the use of fosaprepitant for injection. The majority of severe ISRs, including thrombophlebitis and vasculitis, were reported with concomitant vesicant (anthracycline-based) chemotherapy administration, particularly when associated with extravasation. Necrosis was also reported in some patients with concomitant vesicant chemotherapy. Most ISRs occurred with the first, second or third exposure to single doses of fosaprepitant for injection and in some cases, reactions persisted for two weeks or longer. Treatment of severe ISRs consisted of medical, and in some cases surgical, intervention.

Avoid infusion of fosaprepitant for injection into small veins or through a butterfly catheter. If a severe ISR develops during infusion, discontinue the infusion and administer appropriate medical treatment.

Decrease in INR with Concomitant Warfarin

Coadministration of fosaprepitant with warfarin, a CYP2C9 substrate, may result in a clinically significant decrease in the International Normalized Ratio (INR) of prothrombin time. Monitor the INR in patients on chronic warfarin therapy in the 2-week period, particularly at 7 to 10 days, following initiation of fosaprepitant with each chemotherapy cycle.

Risk of Reduced Efficacy of Hormonal Contraceptives

Upon coadministration with fosaprepitant, the efficacy of hormonal contraceptives may be reduced during administration of and for 28 days following the last dose of fosaprepitant. Advise patients to use effective alternative or back-up methods of contraception during treatment with fosaprepitant and for 1 month following administration of fosaprepitant.

4.5 Interaction with other medicinal products and other forms of interaction

Effect of Fosaprepitant/Apreptant on the Pharmacokinetics of Other Drugs

When administered intravenously, fosaprepitant, a prodrug of apreptant, is converted to apreptant within 30 minutes. Therefore, drug interactions following administration of fosaprepitant for injection are likely to occur with drugs that interact with oral apreptant.

Fosaprepitant, given as a single 150-mg dose, is a weak inhibitor of CYP3A4, and the weak inhibition of CYP3A4 continues for 2 days after single dose administration. Single dose fosaprepitant does not induce CYP3A4. Apreptant is a substrate, an inhibitor, and an inducer of CYP3A4. Apreptant is also an inducer of CYP2C9.

Some substrates of CYP3A4 are contraindicated with fosaprepitant. Dosage adjustment of some CYP3A4 and CYP2C9 substrates may be warranted, as shown in Table.

Table: Effects of Fosaprepitant/Apreptant on the Pharmacokinetics of Other Drugs

CYP3A4 Substrates	
<i>Pimozide</i>	
<i>Clinical Impact</i>	Increased pimozide exposure
<i>Intervention</i>	Fosaprepitant is contraindicated
<i>Benzodiazepines</i>	
<i>Clinical Impact</i>	Increased exposure to midazolam or other benzodiazepines metabolized via CYP3A4 (alprazolam, triazolam) may increase the risk of adverse reactions
<i>Intervention</i>	Monitor for benzodiazepine-related adverse reactions.
<i>Dexamethasone</i>	
<i>Clinical Impact</i>	Increased dexamethasone exposure
<i>Intervention</i>	Reduce the dose of oral dexamethasone by approximately 50%
<i>Methylprednisolone</i>	
<i>Clinical Impact</i>	Increased methylprednisolone exposure
<i>Intervention</i>	Reduce the dose of oral methylprednisolone by approximately 50% on Days 1 and 2 for patients receiving HEC and on Day 1 for patients receiving MEC. Reduce the dose of intravenous methylprednisolone by 25% on Days 1 and 2 for patients receiving HEC and on Day 1 for patients receiving MEC.
<i>Chemotherapeutic agents that are metabolized by CYP3A4</i>	
<i>Clinical Impact</i>	Increased exposure of the chemotherapeutic agent may increase the risk of adverse reactions [see <i>Clinical Pharmacology</i> (12.3)].
<i>Intervention</i>	<u>Vinblastine, vincristine, or ifosfamide or other chemotherapeutic agents</u> • Monitor for chemotherapeutic-related adverse reactions. <u>Etoposide, vinorelbine, paclitaxel, and docetaxel</u> • No dosage adjustment needed.
<i>Hormonal Contraceptives</i>	
<i>Clinical Impact</i>	Decreased hormonal exposure during administration of and for 28 days after administration of the last dose of fosaprepitant
<i>Intervention</i>	Effective alternative or back-up methods of contraception (such as condoms and spermicides) should be used during treatment with fosaprepitant and for 1 month following administration of fosaprepitant.
<i>Examples</i>	birth control pills, skin patches, implants, and certain IUDs
CYP2C9 Substrates	
<i>Warfarin</i>	
<i>Clinical Impact</i>	Decreased warfarin exposure and decreased prothrombin time (INR)
<i>Intervention</i>	In patients on chronic warfarin therapy, monitor the prothrombin time (INR) in the 2-week period, particularly at 7 to 10 days, following administration of fosaprepitant with each chemotherapy cycle.
Other	
<i>5-HT₃ Antagonists</i>	
<i>Clinical Impact</i>	No change in the exposure of the 5-HT ₃ antagonist
<i>Intervention</i>	No dosage adjustment needed
<i>Examples</i>	ondansetron, granisetron, dolasetron

Effect of Other Drugs on the Pharmacokinetics of Fosaprepitant/Apreptant

Apreptant is a CYP3A4 substrate. Co-administration of fosaprepitant with drugs that are inhibitors or inducers of CYP3A4 may result in increased or decreased plasma concentrations of apreptant, respectively, as shown in Table

Table: Effects of Other Drugs on Pharmacokinetics of Fosaprepitant/Apreptant

Moderate to Strong CYP3A4 Inhibitors	
<i>Clinical Impact</i>	Significantly increased exposure of apreptant may increase the risk of adverse reactions associated with fosaprepitant.
<i>Intervention</i>	Avoid concomitant use of fosaprepitant
<i>Examples</i>	<u>Moderate inhibitor:</u> diltiazem <u>Strong inhibitors:</u> ketoconazole, itraconazole, nefazodone, troleandomycin, clarithromycin, ritonavir, nelfinavir
Strong CYP3A4 Inducers	
<i>Clinical Impact</i>	Substantially decreased exposure of apreptant in patients chronically taking a strong CYP3A4 inducer may decrease the efficacy of fosaprepitant [see <i>Clinical Pharmacology</i> (12.3)].
<i>Intervention</i>	Avoid concomitant use of fosaprepitant.
<i>Examples</i>	rifampin, carbamazepine, phenytoin

4.6 Fertility, Pregnancy and lactation

Pregnancy:

Risk Summary

There are insufficient data on use of fosaprepitant in pregnant women to inform a drug associated risk. In animal reproduction studies, no adverse developmental effects were observed in rats or rabbits exposed during the period of organogenesis to systemic drug levels (AUC) approximately equivalent to the exposure at the recommended human dose (RHD) of 150 mg. The estimated background risk of major birth defects and miscarriage for the indicated populations is unknown. In the U.S. general population, the estimated background risk of major birth defects and miscarriage in clinically recognized pregnancies is 2 to 4% and 15 to 20%, respectively.

Data

Animal Data

In embryofetal development studies in rats and rabbits, apreptant was administered during the period of organogenesis at oral doses up to 1,000 mg/kg twice daily (rats) and up to the maximum tolerated dose of 25 mg/kg/day (rabbits). No embryofetal lethality or malformations were observed at any dose level in either species. The exposures (AUC) in pregnant rats at 1,000 mg/kg twice daily and in pregnant rabbits at 25 mg/kg/day were approximately equivalent to the exposure at the RHD of 150 mg. Apreptant crosses the placenta in rats and rabbits.

Lactation

Risk Summary

Lactation studies have not been conducted to assess the presence of apreptant in human milk, the effects on the breastfed infant, or the effects on milk production. Apreptant is present in rat milk. The developmental and health benefits of breastfeeding should be considered along with the mother's clinical need for fosaprepitant and any potential adverse effects on the breastfed infant from fosaprepitant or from the underlying maternal condition.

4.7 Effects on ability to drive and use machines

APRITANT IV may have minor influence on the ability to drive and use machines. Dizziness and fatigue may occur following administration of APRITANT IV.

4.8 Undesirable effects

The following clinically significant adverse reactions are described elsewhere in the labeling:

- Hypersensitivity Reactions [see Warnings and Precautions]
- Infusion Site Reactions [see Warnings and Precautions]

Clinical Trials Experience

Because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice. The overall safety of fosaprepitant for injection was evaluated in approximately 1600 adult patients.

Adverse Reactions in Adults for the Prevention of Nausea and Vomiting Associated with MEC

In an active-controlled clinical trial in patients receiving MEC, safety was evaluated in 504 patients receiving a single dose of fosaprepitant for injection in combination with ondansetron and dexamethasone (fosaprepitant dimeglumine regimen) compared to 497 patients receiving ondansetron and dexamethasone alone (standard therapy). The most common adverse reactions are listed in Table.

Table: Most Common Adverse Reactions in Patients Receiving MEC[†]

	Fosaprepitant for injection, ondansetron, and dexamethasone [†] (N=504)	Ondansetron and dexamethasone [‡] (N=497)
fatigue	15%	13%
diarrhea	13%	11%
neutropenia	8%	7%
asthenia	4%	3%
anemia	3%	2%
peripheral neuropathy	3%	2%
leukopenia	2%	1%
dyspepsia	2%	1%
urinary tract infection	2%	1%
pain in extremity	2%	1%

[†]Reported in ≥2% of patients treated with the fosaprepitant dimeglumine regimen and at a greater incidence than standard therapy.

[‡]Fosaprepitant dimeglumine regimen

[§]Standard therapy

Infusion-site reactions were reported in 2.2% of patients treated with the fosaprepitant dimeglumine regimen compared to 0.6% of patients treated with standard therapy. The infusion-site reactions included: infusion-site pain (1.2%, 0.4%), injection-site irritation (0.2%, 0.0%), vessel puncture-site pain (0.2%, 0.0%), and infusion-site thrombophlebitis (0.6%, 0.0%), reported in the fosaprepitant dimeglumine regimen compared to standard therapy, respectively.

Adverse Reactions in Adults for the Prevention of Nausea and Vomiting Associated with HEC

In an active-controlled clinical study in patients receiving HEC, safety was evaluated for 1143 patients receiving a single dose of fosaprepitant for injection compared to 1169 patients receiving the 3-day regimen of oral aprepitant [see Clinical Studies (14, 1)]. The safety profile was generally similar to that seen in the MEC study with fosaprepitant and prior HEC studies with aprepitant. However, infusion-site reactions occurred at a higher incidence in patients in the fosaprepitant group (3.0%) compared to those in the aprepitant group (0.5%). The following additional infusion-site reactions occurred in HEC study and were not reported in the MEC study described above: infusion-site erythema (0.5%, 0.1%), infusion-site pruritus (0.3%, 0.0%), and infusion-site induration (0.2%, 0.1%), reported in the fosaprepitant group compared to the aprepitant group, respectively.

Because fosaprepitant is converted to aprepitant, those adverse reactions associated with aprepitant might also be expected to occur with fosaprepitant for injection. See the full prescribing information for aprepitant capsules for complete safety information regarding studies performed with oral aprepitant.

Postmarketing Experience

The following adverse reactions have been identified during post-approval use of fosaprepitant. Because these reactions are reported voluntarily from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to drug exposure.

4.9 Overdose

There is no specific information on the treatment of overdosage with fosaprepitant or aprepitant.

In the event of overdose, fosaprepitant should be discontinued and general supportive treatment and monitoring should be provided. Because of the antiemetic activity of fosaprepitant, drug-induced emesis may not be effective in cases of fosaprepitant overdosage.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Mechanism of Action

Fosaprepitant is a prodrug of aprepitant and accordingly, its antiemetic effects are attributable to aprepitant. Aprepitant is a selective high-affinity antagonist of human substance P/neurokinin 1 (NK1) receptors. Aprepitant has little or no affinity for serotonin (5-HT3), dopamine, and corticosteroid receptors, the targets of existing therapies for chemotherapy-induced nausea and vomiting (CINV). Aprepitant has been shown in animal models to inhibit emesis induced by cytotoxic chemotherapeutic agents, such as cisplatin, via central actions. Animal and human Positron Emission Tomography (PET) studies with aprepitant have shown that it crosses the blood brain barrier and occupies brain NK1 receptors. Animal and human studies have shown that aprepitant augments the antiemetic activity of the 5-HT3 receptor antagonist ondansetron and the corticosteroid dexamethasone and inhibits both the acute and delayed phases of cisplatin-induced emesis.

Cardiac Electrophysiology

In a randomized, double-blind, positive-controlled, thorough QTc study, a single 200-mg dose of fosaprepitant (approximately 1.3 times the recommended dose) had no effect on the QTc interval.

5.2 Pharmacokinetic properties

Aprepitant after Fosaprepitant Administration

Following administration of a single intravenous 150-mg dose of fosaprepitant, a prodrug of aprepitant administered as a 20-minute infusion to healthy subjects, the mean AUC_{0-∞} of aprepitant was 37.4 (±14.8) mcg•hr/mL and the mean maximal aprepitant concentration (C_{max}) was 4.2 (±1.2) mcg/mL. Plasma concentrations of fosaprepitant are below the limits of quantification (10 ng/mL) within 30 minutes of the completion of infusion.

Distribution

Aprepitant is greater than 95% bound to plasma proteins. The mean apparent volume of distribution at steady state (V_{ss}) was approximately 70 L in humans.

Aprepitant crosses the blood brain barrier in humans

Elimination

Metabolism

Fosaprepitant is converted to aprepitant in *in vitro* incubations with human liver preparations and in S9 preparations from multiple other human tissues including kidney, lung and ileum. Thus, it appears that the conversion of fosaprepitant to aprepitant can occur in multiple extrahepatic tissues in addition to the liver.

Aprepitant undergoes extensive metabolism. *In vitro* studies using human liver microsomes indicate that aprepitant is metabolized primarily by CYP3A4 with minor metabolism by CYP1A2 and CYP2C19. Metabolism is largely via oxidation at the morpholine ring and its side chains. No metabolism by CYP2D6, CYP2C9, or CYP2E1 was detected.

In healthy young adults, aprepitant accounts for approximately 24% of the radioactivity in plasma over 72 hours following a single oral 300-mg dose of [¹⁴C]-aprepitant, indicating a substantial presence of metabolites in the plasma. Seven metabolites of aprepitant, which are only weakly active, have been identified in human plasma.

Excretion

Following administration of a single intravenous 100-mg dose of [¹⁴C]-fosaprepitant to healthy subjects, 57% of the radioactivity was recovered in urine and 45% in feces.

Aprepitant is eliminated primarily by metabolism; aprepitant is not renally excreted. The apparent terminal half-life ranged from approximately 9 to 13 hours.

Specific Populations

Age: Geriatric Population

Following oral administration of a single 125-mg dose of aprepitant on Day 1 and 80 mg once daily on Days 2 through 5, the AUC_{0-24hr} of aprepitant was 21% higher on Day 1 and 36% higher on Day 5 in elderly (65 years and older) relative to younger adults. The C_{max} was 10% higher on Day 1 and 24% higher on Day 5 in elderly relative to younger adults. These differences are not considered clinically meaningful.

Sex

Following oral administration of a single dose of aprepitant, ranging from 40 mg to 375 mg, the AUC_{0-24hr} and C_{max} are 9% and 17% higher in females as compared with males. The half-life of aprepitant is approximately 25% lower in females as compared with males and T_{max} occurs at approximately the same time. These differences are not considered clinically meaningful.

Race/Ethnicity

Following oral administration of a single dose of aprepitant, ranging from 40 mg to 375 mg, the AUC_{0-24hr} and C_{max} are approximately 27% and 19% higher in Hispanics as compared with Caucasians. The AUC_{0-24hr} and C_{max} were 74% and 47% higher in Asians as compared to Caucasians. There was no difference in AUC_{0-24hr} or C_{max} between Caucasians and Blacks. These differences are not considered clinically meaningful.

Renal Impairment

A single 240-mg oral dose of aprepitant was administered to patients with severe renal impairment (creatinine clearance less than 30 mL/min/1.73 m² as measured by 24-hour urinary creatinine clearance) and to patients with end stage renal disease (ESRD) requiring hemodialysis.

In patients with severe renal impairment, the AUC_{0-∞} of total aprepitant (unbound and protein bound) decreased by 21% and C_{max} decreased by 32%, relative to healthy subjects (creatinine clearance greater than 60 mL/min estimated by Cockcroft-Gault method). In patients with ESRD undergoing hemodialysis, the AUC_{0-∞} of total aprepitant decreased by 42% and C_{max} decreased by 32%. Due to modest decreases in protein binding of aprepitant in patients with renal disease, the AUC of pharmacologically active unbound drug was not significantly affected in patients with renal impairment compared with healthy subjects. Hemodialysis conducted 4 or 48 hours after dosing had no significant effect on the pharmacokinetics of aprepitant; less than 0.2% of the dose was recovered in the dialysate.

Hepatic Impairment

Fosaprepitant is metabolized in various extrahepatic tissues; therefore hepatic impairment is not expected to alter the conversion of fosaprepitant to aprepitant.

Following administration of a single 125-mg oral dose of aprepitant on Day 1 and 80 mg once daily on Days 2 and 3 to patients with mild hepatic impairment (Child-Pugh score 5 to 6), the AUC_{0-24hr} of aprepitant was 11% lower on Day 1 and 36% lower on Day 3, as compared with healthy subjects given the same regimen. In patients with moderate hepatic impairment (Child-Pugh score 7 to 9), the AUC_{0-24hr} of aprepitant was 10% higher on Day 1 and 18% higher on Day 3, as compared with healthy subjects given the same regimen. These differences in AUC_{0-24hr} are not considered clinically meaningful. There are no clinical or pharmacokinetic data in patients with severe hepatic impairment (Child-Pugh score greater than 9).

Body Mass Index (BMI)

For every 5 kg/m² increase in BMI, AUC_{0-24hr} and C_{max} of aprepitant decrease by 9% and 10%. BMI of subjects in the analysis ranged from 18 kg/m² to 36 kg/m². This change is not considered clinically meaningful.

Drug Interactions Studies

Fosaprepitant, given as a single 150-mg dose, is a weak inhibitor of CYP3A4, with no evidence of inhibition or induction of CYP3A4 observed on Day 4. The weak inhibition of CYP3A4 continues for 2 days after single dose administration of fosaprepitant. Aprepitant is a substrate, an inhibitor, and an inducer of CYP3A4. Aprepitant is also an inducer of CYP2C9. Fosaprepitant or aprepitant is unlikely to interact with drugs that are substrates for the P-glycoprotein transporter.

Effects of Fosaprepitant/Aprepitant on the Pharmacokinetics of Other Drugs

CYP3A4 Substrates

Midazolam: Fosaprepitant 150 mg administered as a single intravenous dose on Day 1 increased the AUC_{0-∞} of midazolam by approximately 1.8-fold on Day 1 and had no effect on Day 4 when midazolam was coadministered as a single oral dose of 2 mg on Days 1 and 4.

Corticosteroids:

Dexamethasone: Fosaprepitant administered as a single 150 mg intravenous dose on Day 1 increased the AUC_{0-24hr} of dexamethasone, administered as a single 8-mg oral dose on Days 1, 2, and 3, by approximately 2-fold on Days 1 and 2.

Methylprednisolone: When oral aprepitant as a 3-day regimen (125-mg/80-mg/80-mg) was administered with intravenous methylprednisolone 125 mg on Day 1 and oral methylprednisolone 40 mg on Days 2 and 3, the AUC of methylprednisolone was increased by 1.34-fold on Day 1 and by 2.5-fold on Day 3.

Chemotherapeutic agents:

Docetaxel: In a pharmacokinetic study, oral aprepitant administered as a 3-day regimen (125 mg/80-mg/80-mg) did not influence the pharmacokinetics of docetaxel.

Vinorelbine: In a pharmacokinetic study, oral aprepitant administered as a 3-day regimen (125 mg/80-mg/80-mg) did not influence the pharmacokinetics of vinorelbine to a clinically significant degree.

Oral contraceptives: When oral aprepitant was administered as a 3-day regimen (125-mg/80-mg/80 mg) with ondansetron and dexamethasone, and coadministered with an oral contraceptive containing ethinyl estradiol and norethindrone, the trough concentrations of both ethinyl estradiol and norethindrone were reduced by as much as 64% for 3 weeks post-treatment.

CYP2C9 substrates (Warfarin, Tolbutamide):

Warfarin: A single 125-mg dose of oral aprepitant was administered on Day 1 and 80 mg/day on Days 2 and 3 to subjects who were stabilized on chronic warfarin therapy. Although there was no effect of oral aprepitant on the plasma AUC of R(+) or S(-) warfarin determined on Day 3, there was a 34% decrease in S(-) warfarin trough concentration accompanied by a 14% decrease in the prothrombin time (reported as International Normalized Ratio or INR) 5 days after completion of dosing with oral aprepitant.

Tolbutamide: Oral aprepitant, when given as 125 mg on Day 1 and 80 mg/day on Days 2 and 3, decreased the AUC of tolbutamide by 23% on Day 4, 28% on Day 8, and 15% on Day 15, when a single dose of tolbutamide 500 mg was administered prior to the administration of the 3-day regimen of oral aprepitant and on Days 4, 8, and 15. This effect was not considered clinically important.

Other Drugs

P-glycoprotein substrates: Aprepitant is unlikely to interact with drugs that are substrates for the P-glycoprotein transporter, as demonstrated by the lack of interaction of oral aprepitant with digoxin in a clinical drug interaction study.

5-HT3 antagonists: In clinical drug interaction studies, aprepitant did not have clinically important effects on the pharmacokinetics of ondansetron, granisetron, or hydrodolasetron (the active metabolite of dolasetron).

Effect of Other Drugs on the Pharmacokinetics of Fosaprepitant/Aprepitant

Rifampin: When a single 375-mg dose of oral aprepitant was administered on Day 9 of a 14-day regimen of 600 mg/day of rifampin, a strong CYP3A4 inducer, the AUC of aprepitant decreased approximately 11-fold and the mean terminal half-life decreased approximately 3-fold.

Ketoconazole: When a single 125-mg dose of oral aprepitant was administered on Day 5 of a 10-day regimen of 400 mg/day of ketoconazole, a strong CYP3A4 inhibitor, the AUC of aprepitant increased approximately 5-fold and the mean terminal half-life of aprepitant increased approximately 3-fold.

Diltiazem: In a study in 10 patients with mild to moderate hypertension, administration of 100 mg of fosaprepitant as an intravenous infusion with 120 mg of diltiazem, a moderate CYP3A4 inhibitor administered three times daily, resulted in a 1.5-fold increase in the aprepitant AUC and a 1.4-fold increase in the diltiazem AUC.

When fosaprepitant was administered with diltiazem, the mean maximum decrease in diastolic blood pressure was significantly greater than that observed with diltiazem alone [24.3 ± 10.2 mm Hg with fosaprepitant versus 15.6 ± 4.1 mm Hg without fosaprepitant]. The mean maximum decrease in systolic blood pressure was also greater after co-administration of diltiazem with fosaprepitant than administration of diltiazem alone [29.5 ± 7.9 mm Hg with fosaprepitant versus 23.8 ± 4.8 mm Hg without fosaprepitant]. Co-administration of fosaprepitant and diltiazem; however, did not result in any additional clinically significant changes in heart rate or PR interval, beyond those changes observed with diltiazem alone.

Paroxetine: Coadministration of once daily doses of oral aprepitant 170 mg, with paroxetine 20 mg once daily, resulted in a decrease in AUC by approximately 25% and C_{max} by approximately 20% of both aprepitant and paroxetine. This effect was not considered clinically important.

6.0 Pharmaceutical properties

6.1 Incompatibilities :

APRITANT IV is incompatible with any solutions containing divalent cations (e.g., Ca²⁺, Mg²⁺), including Hartman's and lactated Ringer's solutions. This medicinal product must not be mixed with other medicinal products except those mentioned in Special precautions for disposal and other handling.

6.2 Special precautions for storage:

Store in a refrigerator (2°C – 8°C).

After reconstitution in primary container with 0.9 % Sodium chloride injection and further infusion with 0.9% sodium chloride Injection USP, based on physical and chemical stability evaluation. The reconstituted final drug solution is stable for 24 hours at ambient room temperature [at or below 25°C (77°F)].

From a microbiological point of view, the medicinal product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2 to 8°C.

6.3 Nature and contents of container :

10 mL tubular (USP) Type I clear glass vial with a 13 mm bromobutyl double slotted rubber stopper and sealed with aluminium seal having polypropylene disc

Pack sizes: 1 vial.

6.4 Special precautions for disposal and other handling:

Fosaprepitant for injection vials must be refrigerated, store at 2°C - 8°C (36°F - 46°F).

The reconstituted final drug solution is stable for 24 hours at ambient room temperature [at or below 25°C (77°F)].

APRITANT IV must be reconstituted and then diluted prior to administration.

Preparation of APRITANT IV 150 mg for intravenous administration:

1. Inject 5 ml sodium chloride 9 mg/ml (0.9 %) solution for injection into the vial. Assure that sodium chloride 9 mg/ml (0.9 %) solution for injection is added to the vial along the vial wall in order to prevent foaming. Swirl the vial gently. Avoid shaking and jetting sodium chloride 9 mg/ml (0.9 %) solution for injection into the vial.

2. Prepare an infusion bag filled with 145 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection (for example, by removing 105 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection from a 250 ml sodium chloride 9 mg/ml (0.9 %) solution for injection infusion bag).

3. Withdraw the entire volume from the vial and transfer it into an infusion bag containing 145 ml of sodium chloride 9 mg/ml (0.9 %) solution for injection to yield a total volume of 150 mL. Gently invert the bag 2-3 times.

The medicinal product must not be reconstituted or mixed with solutions for which physical and chemical compatibility has not been established (see section 6.2).

The appearance of the reconstituted solution is the same as the appearance of the diluent.

The reconstituted and diluted medicinal product should be inspected visually for particulate matter and discoloration before administration.

No special requirements for disposal

7. Manufacturer

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9. Date of revision of the text:

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